Journal of Business Research 178 (2024) 114657



Contents lists available at ScienceDirect

Journal of Business Research

journal homepage: www.elsevier.com/locate/jbusres





Virtual versus human: Unraveling consumer reactions to service failures through influencer types

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ARTICLE INFO

Keywords:
Virtual influencer
Service failure
Mind perception
Responsibility attribution
Familiarity

ABSTRACT

In an era where both human and virtual influencers play pivotal roles in shaping consumer behavior, understanding their influence on consumers' reactions to service failures has become an essential scholarly pursuit. This research stands at the forefront of this burgeoning field, dissecting the unique impact of human versus virtual influencers on consumer attitudes toward service failures, including forgiveness propensity, behavioral responses, and punishment intention. Through a series of experimental studies, the research uncovers a nuanced preference for virtual influencers, who elicit higher levels of forgiveness, more positive behavioral responses, and lower punitive intentions during service failures. These findings are mediated by mind perception and responsibility attribution, and are moderated by familiarity, thereby unveiling a complex interplay of psychological processes. The study's contributions transcend mere observations, offering a theoretical framework with empirical testing that integrates consumer psychology with the emergent phenomenon of virtual influencers. The insights gleaned from this research not only enhance academic understanding of the dynamics between influencer types and consumer reactions but also provide actionable strategies for marketers. By bridging theory and practice, this study carves a new path in influencer marketing research, particularly illuminating how to leverage influencer types to mitigate negative consumer reactions due to service failures.

1. Introduction

In recent years, leading consumer brands have increasingly turned to virtual influencers to provide customized, scandal-free social media content (Qu & Baek, 2023; Wu et al., 2023). The emergence of various types of influencers, including human, virtual influencers and digital avatar, has revolutionized brand communication and consumer engagement (Audrezet & Koles, 2023; Nghiêm-Phú & Suter, 2023). For instance, beyond their outward appearance, digital avatars used in digital humanoids are more capable of eliciting the needed attractiveness and curiosity through their tailored content (Foster et al., 2022). Brands can customize virtual influencers or digital avatars in terms of their appearance and the manner in which they present a product (Yao et al., 2023), thereby strengthening consumer-brand relationships (Elsharnouby et al., 2023; Foster et al., 2022). Despite the growing body of research on how influencers can be used for interactive marketing strategies and brand identity enhancement (Wang, 2021; 2023; Audrezet & Koles, 2023), the nuanced differences between the different types of influencers remain underexplored.

The existing literature provides few guidelines regarding situations where negative effects occur due to influencers' endorsement of a faulty product or the provision of incorrect advice (Folkes et al., 1987; Cleeren et al., 2017). There is still much to learn about whether and how consumer perceptions differ for different types of influencers in terms of the

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https://doi.org/10.1016/j.jbusres.2024.114657
Received 2 February 2023; Received in revised form 29 March 2024; Accepted 8 April 2024
Available online 3 May 2024
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mind perception (Ryding et al., 2023; Waytz et al., 2010) and other attributes. Additionally, the attribution of responsibility for service failures between different types of influencers is not well understood. The ambiguity in this area calls for further research to understand the psychological mechanisms underpinning potential differences and to contribute to the burgeoning field of influencer marketing literature with new insights (Wang, 2022). This study attempts to build theoretical perspectives upon the different types of influencers to understand consumers' psychological responses to service failures. The integration of perspectives from mind perception (Waytz et al., 2010; Tzelios et al., 2022) and attribution theory (Folkes, 1984) allows for a multifaceted exploration of the complex interplay between influencers and consumers. By exploring differences in consumers' attributions and attitudes toward service failure based on different types of influencers, this study seeks to advance our understanding of the types of influencers and their subtle influence on consumers in circumstances of service failure. The research findings offer theoretical and managerial insights into influencer selection and strategies for consumer-brand relationship repair, thereby enriching both academic discourse and practical applications.

The study is structured as follows: first, the theoretical background of the research is presented, focusing on the types of influencers, leading to the development of the hypotheses. The subsequent sections report the results of four experimental studies that explore the differences in consumers' attributions and attitudes toward service failure based on different types of influencers, and the moderating role of familiarity. The studies provide a comprehensive exploration of the effects of influencer types and delve into underlying psychological mechanisms. Finally, theoretical and practical implications are discussed, along with directions for future research in influencer marketing, especially in the context of service failures.

2. Theoretical background and hypotheses

2.1. Virtual influencers vs. human influencers

A virtual influencer, defined as an interactive, real-time character—whether humanlike or not—controlled and presented by artificial intelligence to consumers in a digital environment, has emerged as a significant phenomenon (Sands et al., 2022c). These virtual influencers, exhibiting characteristics akin to human behavior (Seymour et al., 2019; Wang & Huang, 2022), possess distinct personalities and growth stories (Thomas & Fowler, 2021; Araque & Moynihan, 2021). Enabled by digital technologies, virtual influencers have the capacity to interact with followers in a personalized and real-time manner (Campbell et al., 2020, 2021), a feature that has rendered them attractive to online users. Notably, engagement with virtual influencers on platforms such as Instagram is almost three times higher than with human influencers (Baklanov, 2019), with successful virtual influencers like Lu do Magalu accumulating millions of followers across various social media platforms (Sands et al., 2022c).

The commercial appeal of virtual influencers has not gone unnoticed by companies, leading to an increase in their use for brand promotion (Sands et al., 2022c; Thomas & Fowler, 2021). Prominent virtual influencers, such as Lil Miquela, have collaborated with high-profile brands, reflecting their considerable market value (Time Staff, 2018). Meanwhile, some companies have pioneered the creation of their own virtual influencers, as exemplified by Prada's virtual influencer, Candy (Laing, 2020).

In anticipation of the potential trend of virtual influencers gradually replacing human influencers in the commercial field, researchers have embarked on comparative studies to discern the advantages and disadvantages of virtual versus human influencers. While some studies report no significant differences in aspects such as intention to follow, perceived personalization, and viewers' parasocial responses (Sands et al., 2022a; Stein et al., 2022), others highlight the drawbacks of virtual influencers, suggesting that virtual influencers are less effective as endorsers in terms of brand attitude and purchase intention (Ozdemir et al., 2023), which is attributed to a perceived lower transparency (Lopez & Garza, 2023) or a lack of trust and credibility (Sands et al., 2022a; Ozdemir et al., 2023). The phenomenon is further analyzed through the lens of the uncanny valley theory, where human-like virtual influencers have been observed to receive fewer positive reactions compared to human influencers (Arsenyan and Mirowska, 2021). Despite potential advantages such as evoking word-of-mouth intentions (Sands et al., 2022a) and leading to higher perceived ad novelty (Franke et al., 2022), the specific advantages of virtual influencers over human ones in the context of service failure remain a largely unexplored area.

2.2. Service failure by virtual vs. human influencers

Influencers, both human and virtual, frequently encounter service failures, such as recommending faulty products, providing incorrect advice, or expressing opinions offensive to their followers (Kubbernus, 2020). Virtual influencers are not immune to such failures, leading to potential relationship crises. A case in point is the virtual influencer Lil Miquela's controversial discussion on YouTube about a fictional harassment experience, which caused offense among her followers (Idika, 2019). With service failures becoming an increasingly common occurrence, recent studies have explored influencers' response strategies (Hudders et al., 2022), types of transgressions (Cocker et al., 2021), and the effects of influencer betrayal (Reinikainen et al., 2021). Yet, the extant literature has largely overlooked how consumer attitudes and behaviors differ when encountering service failures caused by different influencers, namely human vs. virtual influencers. While the literature regarding service failures in the virtual influencer research is still in the nascent stage, theoretical insights can be gleaned from studies comparing the service effectiveness of artificial intelligence servicers (e.g., AI Chatbot and virtual service agent) to their human counterparts. This comparison is particularly relevant as consumers may regard virtual influencers as algorithms, thereby influencing their behavioral responses (Dietvorst et al., 2015).

Existing literature suggests the existence of a phenomenon known as algorithm aversion, where consumers prefer human advice over algorithms, despite algorithms often outperforming human judgment (Dietvorst et al., 2015; Castelo et al., 2019). This aversion extends to various domains, from healthcare (Longoni et al., 2019) to restaurant services (Mende et al., 2019), reflecting concerns over the potential threat to human identity or a lack of empathy (Wu et al., 2023). Meanwhile, other studies have revealed lower consumer satisfaction and purchase intention following a service failure with a virtual agent compared to a human agent (Sands et al., 2022b). In many contexts, consumers tend to have low mind perception of non-human algorithms, recognizing their superiority in computing power and accuracy (Himmelstein & Budescu, 2022; Reich et al., 2022). In such a context, human minds are often perceived as more flexible and emotional, making it critical for social media influencers to interact with followers in a nuanced and adaptable manner (Saldanha et al., 2023), considering that the expectations tend to gravitate towards abilities traditionally associated with the human mind rather than artificial intelligence or digital avatar (Qu & Baek, 2023).

The literature is not univocal in its assessment of AI in the context of service failure. Some studies have demonstrated advantages of AI, such as consumers being less prone to share negative word-of-mouth after an AI-induced service failure (Huang & Philp, 2021) or showing more forgiveness towards AI-controlled autonomous vehicles in traffic accidents (Awad et al., 2020; Gill, 2020). Such divergent streams of research do not yield a consistent conclusion regarding the relative merits and demerits of AI and human service providers. The contrasting effects between virtual and human influencers in the context of service failures, therefore, merit further investigation. In light of the frequent occurrence of service failures in the social media influencer business and the lack of

comparative research between virtual and human influencers, this study aims to explore consumers' reactions to service failures caused by virtual vs. human influencers.

2.3. The role of mind perception in service setting

This research primarily builds upon the mind perception theory to investigate the impact of different types of influencer on consumer perception and attitude in the context of service failure. Mind perception, also known as humanizing or mentalizing, refers to the human cognition process, which is capable of extricating information and inferring about others' goals, beliefs, feelings and intentions (Gray et al., 2007; Tzelios et al., 2022). In a nutshell, mind perception involves how people perceive the minds of other entities (humans, non-human animals, and non-living objects and forces (Waytz et al., 2010). Despite the fact that inanimate objects like virtual influencers are mindless machines, consumers nevertheless routinely attribute humanlike mental capacities to them in their perception and judgment (Epley, 2018). Accordingly, mind perception has a direct impact on consumer response to virtual influencers or digital avatars.

A comprehensive review of the existing literature reveals that consumers' aversion to AI service is primarily evident in tasks that are perceived to require human intellect for optimal execution (Castelo et al., 2019). Several studies have shed light on this phenomenon, identifying perceptions of AI as less empathetic and knowledgeable compared to human agents (Luo et al., 2019), or attributing a negative brand attitude and reduced purchase intention to virtual influencers due to their perceived lack of sensory capability and credibility (Ozdemir et al., 2023).

Interestingly, the literature also unveils a spectrum of consumer reactions guided by perceptions of AI's deficiency in human-like mental characteristics, often based on stereotyped perception (Au & Ng, 2020). This includes a reduced sense of guilt and a greater propensity towards unethical consumer behaviors when interacting with non-human agents (Kim et al., 2023a), or a focus on how artificial agents serve humans rather than on their underlying intentions (Kim & Duhachek, 2020). On the other hand, positive effects have been observed when AI agents are endowed with human-like traits. Such effects range from increased customer satisfaction with more human-like virtual agents (Söderlund and Oikarinen, 2021), to favorable consumer reactions to humanoid robots involved in service failure (Choi et al., 2020), and enhanced perceptions of credibility and quality with human-like endorsements and capabilities (Kim et al., 2023b; Söderlund, 2022a; 2022b; Upadhyay and Kamble, 2023). Foster et al. (2022) suggest that the social functions provided by the digital avatar are more likely to facilitate the consumer–brand relationship (CBR) process than transactional functions. Recently, Yao et al. (2023) find that in the context of a customer service failure, an anthropomorphized intelligent customer service avatar that appeared competent (vs. warm) induced greater customer disappointment. However, if the anthropomorphic avatar had a cartoon-like appearance, the effect of avatar image perception (competent vs. warm) on consumers' willingness to interact diminished. These findings collectively underscore that consumers' belief in AI service providers possessing human-like minds significantly influences their attitudes towards their services.

2.4. Mind perception and attribution of service failure by influencers

While existing literature underscores the role of mind perception in AI services and virtual influencers, this effect may manifest differently in the context of service failure. For instance, Garvey et al. (2023) demonstrated that in better-than-expected service scenarios, consumers favor humans over AI agents due to perceived benevolent intentions. Conversely, in worse-than-expected situations, consumers respond more favorably to AI agents, perceived as unintentional and thus not accountable for poor service (Srinivasan and Sarial-Abi, 2021). Virtual influencers, despite being AI-driven, possess unique characteristics such as a more realistic human image, personalized settings, and closer fan interactions (Wang & Huang, 2022; Thomas & Fowler, 2021), which distinguish them from other AI service providers. This study aims to explore consumer attitudes towards service failures of virtual versus human influencers, grounded in mind perception and attribution theories.

Besides mind perception, attribution theory literature further reveals that consumer reactions to service failures hinge on their interpretations of the failures (Folkes, 1984). Anger, regret and retribution ensue when consumers perceive the service provider as responsible, especially if the harm was avoidable (Folkes et al., 1987; Cleeren et al., 2017; Wu and Wang, 2017). Building upon theoretical work on moral attribution, humans assign moral responsibilities only to agents perceived to have sufficiently high levels of mind (Lanz et al., 2023; Smith et al., 2023). Consequently, higher mind perception corresponds to greater accountability for harm (Waytz et al., 2014), while in the business context, AI agents often lead to less blame and enhanced forgiveness (Arikan et al., 2023).

In light of these insights, this study posits that consumers' evaluations of service failures by influencers may differ depending on whether the influencer is an algorithm-driven virtual entity or a real-life human. Virtual influencers, potentially perceived as possessing less human-like mind, may be attributed less responsibility for service failures, engendering a more favorable attitude towards their transgressions. This study will empirically test the consequences of consumers' attitudes towards influencer service failure in terms of three dependent measures: (1) forgiveness propensity, which is a common metric to gauge consumers' attitudes towards service failure (Kim et al., 2023; Sinha & Lu, 2016); (2) behavioral response such as reluctance to purchase recommended products (Yang & Hu, 2021; Kubbernus, 2020); and (3) punishment intention, including complaints and dissemination of negative word-of-mouth (Huang & Philp, 2021; López-López et al., 2021; Gregoire & Fisher, 2008).

We argue that mind perception and responsibility attribution mediate the impact of influencer type on three outcome measures of service failure. Specifically, in the process of interacting with the two types of influencers, consumers may feel that virtual influencers have lower human mind than the human ones, which makes them attribute less accountability to virtual influencers (Garvey et al., 2023; Zhu et al., 2023). As a result, the lower mind perception tend to reduce the likelihood to attribute service failures to virtual influencers' responsibility (Smith et al., 2023), leading to a more forgiving attitude (Arikan et al., 2023). When consumers adopt a more forgiving attitude, they are also more tolerant and willingness to offer opportunities for recovery and other compensatory massacres, rather than immediately taking negative behaviors such as reluctance to purchase recommended products. In the same vein, a more forgiving attitude also counteracts consumers' negative feelings like anger, making them more lenient and meanwhile inhibiting their urge to take punitive actions like complaints and dissemination of negative word-of-mouth (Huang & Philp, 2021). Based on the above logic, the following hypotheses are proposed.

H1: The type of influencers (human vs. *virtual) affects consumers' responses to service failures.*

H1a: Consumers show a higher forgiveness propensity towards the service failures of virtual influencers than human ones.

H1b: Consumers show more positive behavioral responses towards the service failures of virtual influencers than human ones.

H1c: Consumers show a lower punishment intention towards the service failures of virtual influencers than human ones.

*H2: Consumers' attitudes towards service failures influenced by virtual influencers (*vs. *human influencers) are sequentially mediated first by mind perception and then by responsibility attribution..*

2.5. Agency or experience?

Mind perception delineates two core components: agency, reflecting an entity's perceived capacity for intention and action (e.g., self-control, judgment, communication); and experience, signifying the entity's perceived capacity for sensation and feeling (e.g., hunger, fear, pain) (Gray & Wegner, 2012). When the role of agency is considered, human influencers, compared to AI influencers, engage in more immediate and intimate interaction with followers, demanding flexibility in service provision and activity initiation (Ouvrein et al., 2021; Yim et al., 2023). These traits may lead to elevated consumer expectations for influencer agency. Consequently, consumers may attribute service failure to virtual influencers' low levels of agency, exhibiting understanding and forgiveness, while attributing blame to human influencers. On the other hand, since influencers often make product recommendations based on their experience (Leite and Baptista, 2022), they are supposed to be capable of accurately perceiving followers' emotions and needs for effective communication (Holiday et al., 2023), increasing consumers' expectations for influencers' experiential abilities. In this regard, AI-based virtual influencers are considered to lack such capabilities compared to human influencers (Longoni & Cian, 2022; Wien & Peluso, 2021), leading to consumer attribution of service failures.

Given the focus of the present study is on the comparison of specific characteristics between different types of influencers, both agency and experience are considered as sub-dimensions of mind perception that mediate the influence of influencer types on consumers' attitudes towards service failure. The extant literature is unclear which sub-dimension of mind perception, i.e., experience or agency, can better explain such an effect. Previous studies have predominantly focused on the experience dimension of mind perception to elucidate the disparities between AI and human service providers. For instance, Yam et al. (2021) and Sullivan and Wamba (2022) have emphasized the role of experience in customer satisfaction and blame judgments and Gray et al. (2007) articulated that the differentiation between artificial intelligence (AI) and humans predominantly lies in the dimension of experience rather than agency. While AI may demonstrate proficiency in agency, such as the ability to plan and act, their capacity to experience conscious emotions and senses (e.g., shame, pride, sympathy, pain) is often questioned (Epley & Waytz, 2010). However, the role of agency should not be ignored, as agency may also contribute to perceived humanness and customer satisfaction (Söderlund and Oikarinen, 2021; Srinivasan and Sarial-Abi, 2021). In line with the above discussions, we further test two alternative hypotheses by comparing whether agency or experience plays a more important mediating role by employing a parallel mediation model in subsequent empirical studies.

2.5.1. The moderating role of familiarity

While the preceding rationale suggests that in the context of service failure, the human mind might negatively influence consumers' attitudes, this effect could be contingent on specific conditions. For instance, familiarity might moderate such a relationship because familiarity has been found to influence customer satisfaction, repurchase intentions, and word-of-mouth intentions (Hoyer & Brown, 1990; Söderlund, 2002).

It has been well documented in the literature that familiarity tends to increase liking and reduce perceived risks towards humans and objects (Bornstein, 1989; Munoz and Towner, 2022). Research shows that familiarity can mitigate the negative impacts of service failure and brand crises (Kim et al., 2022; Dawar & Lei, 2009; Brady et al., 2008). For instance, familiar restorers are more likely to enhance restoring satisfaction after a service failure (Nazifi et al., 2021), as they are less evaluated based on the failure alone (Pham & Muthukrishnan, 2002). Some studies have also identified the positive effect of familiarity in diminishing algorithm aversion (Castelo et al., 2019; Berger et al., 2021), as it can clarify issues due to misunderstandings and biased opinions about AI (Gonzalez et al., 2022). In their investigation of human belief in the mental lives of nonhuman animals (particularly how and why people attribute emotions to other species), Morris et al. (2012) demonstrate that familiarity with animals plays an important role in determining beliefs about emotions in animals and animal mind. However, existing literature has not thoroughly explored whether the role of familiarity differs between human and virtual service providers or influencers. As such, the present research examines the interaction effect between familiarity and influencer type, with a focus on affective reactions and the characteristics of the human mind.

Psychological research has highlighted the positive association between familiarity and affective reactions that result in positive attitudes (Monin, 2003; Patterson & Mattila, 2008). Affection, a feature typically ascribed to the human mind, is often perceived as lacking in algorithms (Gray et al., 2007; 2012). We contend that human influencers, due to the advantages of the human mind, may foster deeper affective connections with followers during the process of increasing familiarity, thereby enhancing consumer forgiveness for service failures. In contrast, virtual influencers may be less effective in building such a two-way affection connection and mutual familiarity, rendering the positive effect of familiarity less significant on attitudes towards service failure. Accordingly, the following hypothesis is proposed:

H3: Familiarity moderates the effect of influencer types on consumers' attitudes towards service failures. In particular, familiarity enhances consumers' attitudes towards service failures of human influencers more than virtual influencers.

2.6. Overview of the studies

Based on the above hypotheses, we propose the theoretical framework as depicted in Fig. 1. The experimental method is used for the present research for two primary reasons. First, it is currently the most reliable method for testing causality between variables. Second, the effective control of extraneous variables makes experimental design conducive to constructing scenarios that create a clean comparison between virtual and human influencers. This is especially important given the difficulty in finding real-world datasets describing situations in which a virtual and human influencer suffer from identical service failures. Zhou et al. (2023) provided a comprehensive review of the findings and methodologies in virtual influencer research. Their work underscores the prevalence of experimental scenarios in comparing virtual and human influencers in existing research. For instance, Ozdemir et al. (2023) implemented experimental methods where participants viewed images of brands endorsed by either human or virtual influencers, followed by an assessment of their brand attitudes. In alignment with preceding research (e.g., Zhou et al., 2023; Ozdemir et al., 2023; Zhou et al., 2024), The present study leverages the experimental method. We manipulate the type of influencer by presenting participants with images and descriptive narratives of either virtual or human influencers, subsequently evaluating their responses to the influencers' service failures. Each empirical study is organized as follows. We first conduct two pilot studies, using various experimental materials and scenarios to test the robustness of the main effects (H1) and to rule out the potential effects caused by confounding factors: service failure types and influencers' gender (see the Web Appendix for the results). Following this, Study 1 mainly focuses on exploring the explanatory mechanism of the main effect. We examine the mediating roles of the two sub-dimensions of mind perception (i.e., agency and experience) and simultaneously rule out alternative explanations of perceived cuteness and perceived novelty, using the serial mediation model with mind perception and responsibility attribution (H2). Finally, Study 2 concentrates on exploring the boundary condition of familiarity (H3).

3. Pilot studies

Pilot-study 1 employs authentic images and descriptive narratives of both virtual and human influencers as the experimental stimuli (details

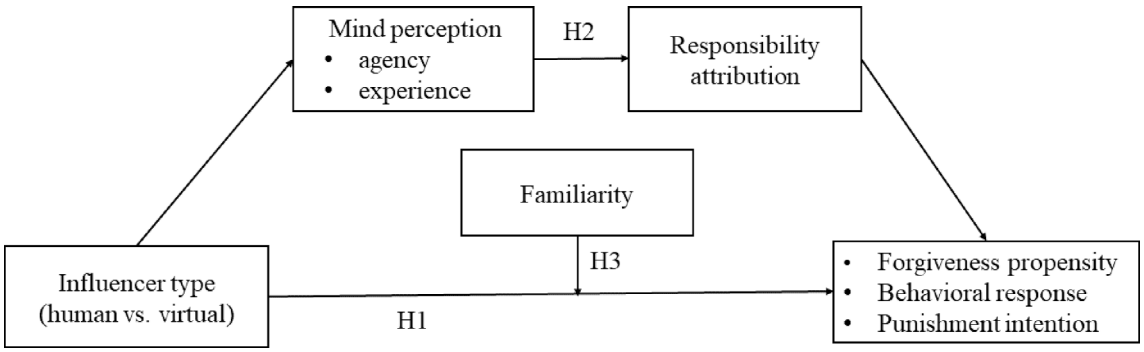


Fig. 1. Theoretical framework.

available in the Web Appendix).. This approach allows us to check whether consumers consistently demonstrate a more favorable attitude (measured by higher forgiveness propensity, more positive behavioral response and lower intention to take punitive actions) towards the service failures of virtual influencers compared to their human counterparts, as predicted in H1. In this regard, we aim to demonstrate that such an effect holds steady across various service failure scenarios, including product failure, recommendation failure, and value-based failure (for statistical analysis methodologies, refer to the Web Appendix).

We also conduct Pilot study 2 with a variation by using fictitious images of influencers to mitigate potential confounding variables due to differences in the images or participants' prior experience with the influencers. This study confirmed H1 and additionally ruled out the effect due to gender difference. Thus, the main effect is found to be consistent across both male and female influencers, further bolstering the foundational hypothesis for the subsequent main studies. Since our primary focus in the present research is on testing the underlying mechanisms and boundary conditions of the relationship, the details of the results from the pilot studies are reported in the appendix (see the Web Appendix for details).

4. Study one

4.1. Research design and participants

Study 1 is designed with four objectives. Firstly, it explores the mediating role of mind perception, focusing on its two sub-dimensions (i.e., agency and experience). Secondly, the study aims to eliminate potential alternative explanations by examining the mediation of perceived cuteness and perceived novelty based on mind perception. Thirdly, the study tests a serial mediation model with mind perception and responsibility attribution as mediators. A single factor between-subjects design is employed, distinguishing human from virtual influencers. Lastly, we test alternative hypotheses by comparing the mediating effects of agency and experience.

4.2. Procedure

A total of 200 participants (M_{age} = 31.63, SD = 8.15, 42.0 % males) were recruited via Credamo (a data collection platform in China) and randomly assigned to the two conditions.

The procedure for Study 1 mirrored that of the pilot. Participants are initially provided with a brief description of the influencer, accompanied by an image (detailed in the Web Appendix). As Pilot-study 2 has established that the gender of the influencer did not have a significant effect, and considering that most influencers are currently female, Study 1 utilized the same female images as those used in Pilot-study 2 as experimental stimuli (see Fig. 2). To validate the image materials and rule out alternative explanations, participants' perceived attractiveness, perceived colorfulness, familiarity, and trust in the influencers are

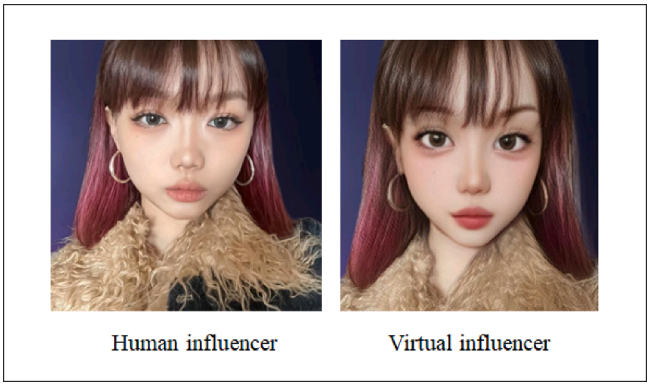


Fig. 2. Experimental stimuli of study 1.

measured. Additionally, participants are asked to indicate to what extent they perceive the influencers to look like an influencer, a human or a virtual influencer (see Web Appendix).

The same scenario of product failure in Pilot-study 1 is employed in Study 1, which describes the influencer recommending an exclusive coffee cup with exquisite design, but the followers find the quality to be poor and far below expectations. Next, participants are prompted to indicate their forgiveness (Cronbach's α = 0.93), behavioral responses (Cronbach's α = 0.94), and punishment intention (Cronbach's α = 0.92), consistent with that found in pilot studies (see Web Appendix of Pilot-study 1). Participants also report their mind perception of the influencer and infer the responsibility attribution.

Mind perception is measured by asking participants to rate (on a 5-point scale) the influencer on twelve different mental capacities ("To what extent do you think the influencer can…"; 1 = not at all, 5 = extremely; Cronbach's α = 0.96) (Bigman & Gray, 2018). Six are agency-related: "communicate with others," "is able of thinking," "plans his actions," "is intelligent," "has foresight" and "is able to think things through" (Cronbach's α = 0.93), and the other six were experience-related: "sensitive to pain", "experience happiness", "experience fear", "experience compassion", "experience empathy" and "experience guilt" (Cronbach's α = 0.95).

Responsibility attribution is measured by three 7-point Likert-scale items adopted from Wan and Wyer (2019) (Cronbach's α = 0.94; 1 = disagree very much, 7 = agree very much): (I think this problem is attributed to the influencer; I think the influencer should take full responsibility for the problem; and I think the problem is beyond the control of the influencer). An attention check item was included at the end of the survey, wherein participants are informed to identify whether the influencer they are shown is human or virtual. Incorrect responses are automatically excluded from the sample by the Credamo platform.

4.3. Results and discussions

A series of independent samples t-tests were conducted to ensure the validity of the experimental stimuli. The results revealed no significant differences between the virtual and human influencer groups in terms of attractiveness, colourfulness, familiarity, trust, and the likeness to a social media influencer. However, participants in the virtual group perceived the image as less human-like compared to those in the human group. These statistical details can be found in the Web Appendix. Overall, the findings indicate that the experimental materials are appropriate for this study.

Main Effects. A series of independent samples t-tests on the dependent variables showed that participants demonstrated a higher forgiveness propensity ($M_{virtual}$ = 5.00 ± 1.39, M_{human} = 4.05 ± 1.78, $t(198)$ = -4.22, $p < 0.001$, Cohen's $d = 0.59$), a more positive behavioral response ($M_{virtual} = 3.44 \pm 1.60$, $M_{human} = 4.34 \pm 1.84$, $t(198) = 3.72$, $p < 0.001$, Cohen's $d = 0.52$), and a lower punishment intention ($M_{virtual} = 3.27 \pm 1.48$, M_{human} = 4.33 ± 1.64, $t(198)$ = 4.80, $p < 0.001$, Cohen's d = 0.68) towards the service failures of virtual influencers than human influencers (See Fig. 3). H1 is supported.

The Mediating role of mind perception. The influencer type is recoded as a dummy variable (0 = human, 1 = virtual). Using PROCESS macro in SPSS (Hayes & Preacher, 2014), we ran three bootstrap mediation analyses (Model 4) in which the two sub-dimensions of mind perception (i.e., agency and experience) served as parallel mediators and the forgiveness propensity, behavioral response, and punishment intention served as dependent variables respectively. As shown in Table 1, all the mediation effects of agency are statistically significant but none of the mediation effects of experience is significant. These results suggest that perceived agency, compared to perceived experience, is a better explanatory mechanism between influencer type and dependent variables. Thus, in the subsequent statistical analysis, we will only use the sub-dimensions of agency in mind perception scale as the mediator.

In order to figure out whether the main effect can be explained by other mechanisms, we employed parallel mediation models to analyze the mediating role of perceived agency, novelty, and cuteness. Using PROCESS macro in SPSS, we ran three bootstrap mediation analyses (Model 4) in which the perceived agency, novelty, and cuteness served as parallel mediators and the forgiveness propensity, behavioral response, and punishment intention served as dependent variables respectively. As shown in Table 2, both perceived agency and cuteness mediate the main effect on forgiveness propensity. However, in the models where behavioral response and punishment intention served as dependent variables, only perceived agency plays a mediating role.

Table 1
The results of parallel mediation analyses of mind perception.

Dependent Variables	Mediators	Effect	BootSE	BootLLCI	BootULCI
Forgiveness Propensity	Agency	0.49	0.168	0.199	0.864
	Experience	0.33	0.204	−0.115	0.692
Behavioral Response	Agency	−0.67	0.185	−1.060	−0.334
	Experience	−0.13	0.210	−0.531	0.311
Punishment Intention	Agency	−0.47	0.182	−0.870	−0.151
	Experience	−0.33	0.225	−0.736	0.147

Table 2
The results of parallel mediation analyses of alternative explanations.

Dependent Variables	Mediators	Effect	BootSE	BootLLCI	BootULCI
Forgiveness Propensity	Agency	0.74	0.142	0.496	1.048
	Novelty	0.12	0.185	−0.211	0.507
	Cuteness	0.13	0.080	0.005	0.317
Behavioral Response	Agency	−0.79	0.152	−1.112	−0.513
	Novelty	−0.15	0.194	−0.566	0.202
	Cuteness	−0.08	0.073	−0.243	0.045
Punishment Intention	Agency	−0.71	0.141	−1.015	−0.463
	Novelty	−0.14	0.178	−0.520	0.178
	Cuteness	−0.08	0.068	−0.237	0.036

Note. LLCI means the lower level of 95% confident intervals and ULCI means the upper level of 95% confident intervals. If these two interval values do not include 0, the effect is significant at the 0.05 level.

These results suggest that perceived agency is a better explanatory mechanism than other variables (i.e., novelty and cuteness).

Serial Mediation Effect of Mind Perception and Responsibility Attribution. We further test our proposed whole serial mediation models (H2), using Model 6 in PROCESS macro (Hayes & Preacher, 2014). We ran three bootstrap serial mediation analyses in which perceived agency served as the first mediator and responsibility attribution as the second mediator, with forgiveness propensity, behavioral response, and punishment intention as dependent variables. As shown in Table 3, there are significant indirect effects of "perceived agency → responsibility attribution" between influencer type and the dependent variables, suggesting that the serial mediation effects of perceived agency and responsibility attribution explain the difference in the dependent variables between human and virtual influencer (Table 4).

Discussion. Study 1 supports H2, showing that the relationship between influencer type and consumer attitude to service is sequential mediated by mind perception and responsibility attribution. Meanwhile, results from a parallel mediation analysis further reveal that perceived

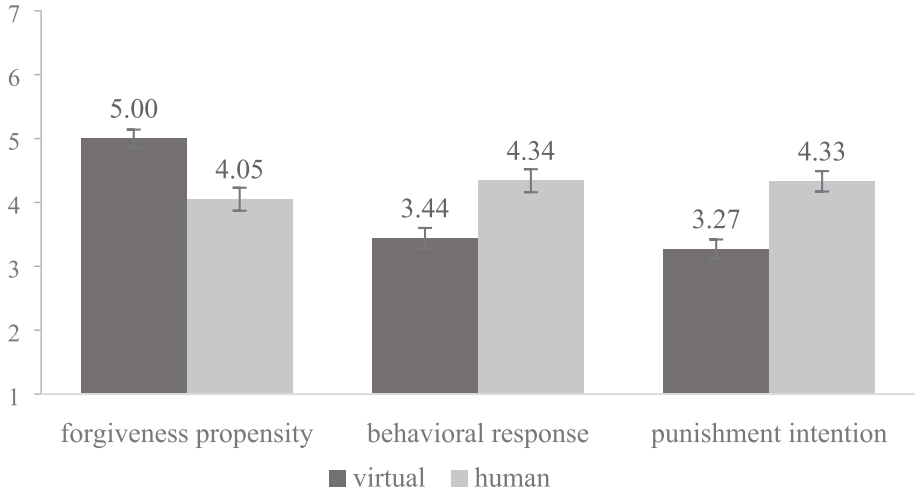


Fig. 3. Difference of dependent variables between virtual and human influencers.

Table 3
The results of serial mediation analyses.

Dependent Variables	Paths	Effect	BootSE	BootLLCI	BootULCI
Forgiveness Propensity	IT → PA → FP	0.35	0.121	0.127	0.604
	IT → RA → FP	0.60	0.171	0.268	0.937
	IT → PA → RA → FP	0.34	0.109	0.153	0.575
Behavioral Response	IT → PA → BR	−0.40	0.129	−0.659	−0.158
	IT → RA → BR	−0.62	0.182	−0.981	−0.264
	IT → PA → RA → BR	−0.35	0.116	−0.601	−0.156
Punishment Intention	IT → PA → PI	−0.31	0.095	−0.501	−0.141
	IT → RA → PI	−0.65	0.188	−1.004	−0.270
	IT → PA → RA → PI	−0.37	0.109	−0.603	−0.178

Note. IT is for influencer type; PA is for perceived agency; RA is for responsibility attribution; FP is for forgiveness propensity; BR is for behavioral response; PI is for punishment intention; LLCI means the lower level of 95% confident intervals and ULCI means the upper level of 95% confident intervals. If these two interval values do not include 0, the effect is significant at the 0.05 level.

agency (vs. perceived experience) serves as a better explanatory mechanism. In addition, Study 1 has ruled out alternative explanatory mechanisms, such as perceived cuteness and novelty, through parallel mediation analyses. Interestingly, Study 1 also suggests that perceived cuteness can explain the main effect of influencer type on forgiveness propensity, which is grounded in perceived agency. In other words, participants are more likely to be forgiving if they perceive the virtual influencer to be cute. This finding aligns with the research by Lv et al. (2021), which demonstrated a greater likelihood of forgiving a cuter virtual image. However, our study does not show that cuteness mediates the effect of influencer type on behavioral response and punishment intention. Given that cuteness primarily describes the appearance rather than the mind of virtual influencers, we argue that perceived agency serves as a more essential and appropriate explanatory mechanism.

5. Study two

5.1. Research design and participants

Study 2 serves a dual purpose. First, considering that the stimuli used for influencers in Study 1 were not real influencers, Study 2 employs real virtual and human influencers from Instagram as experimental materials. It also uses their genuine followers as participants to provide a validity test for H1 and H2. Second, Study 2 tests H3 by examining the moderating role of followers' familiarity. Two types of familiarity are examined: consumers' general familiarity with the concept and phenomenon of virtual or human influencers, and their specific familiarity with a particular virtual or human influencer featured in the experimental materials. Scales measured the former, and the experimental materials manipulate the latter.

Study 2 employs a 2 (influencer type: human vs. virtual) × 2 (familiarity: familiar vs. unfamiliar) between-subjects design. We recruited 308 participants via Prolific and randomly assigned them to one of the four conditions. Seventeen participants who failed the attention check (i.e., identifying whether they were exposed to virtual or human influencer and their familiarity with each one) were removed from the sample. The mean age of the remaining participants is 38.03 ± 12.48, and 45 % of them were males.

5.2. Materials

Study 2 employs real influencers from Instagram as the experimental materials. After selecting five virtual influencers from Instagram for the virtual influencer groups, we selected five human influencers from Instagram who correspond closely with the virtual ones (in terms of follower number, appearance, and areas of influence) for the human influencer groups. Participants are presented with a succinct introduction of the influencers, accompanied by a picture of their profiles on Instagram (details are available in the Web Appendix). They are then asked to choose one of these influencers, in accordance with the familiarity groups to which they were assigned, to proceed with completing the questionnaire.

5.3. Procedure

Instagram users who have recent experience following one or more virtual (vs. human) influencers on Instagram are recruited for the virtual (or human) influencer condition. After being introduced to a brief cover story about virtual (or human) influencers, participants are asked to answer two questions that assess their general familiarity with virtual (or human) influencers: (1) Their familiarity with the concept of a

Table 4
Summary of studies.

Study	Objective	Materials	Findings
Pilot-study 1 in Web Appendix	Examining whether the influencer type (human vs. virtual) can affect consumers' responses to different types of service failure (H1).	Choosing a real AI influencer *Higaga* introduced by Baidu as the virtual influencer and a real human influencer who looks like *Higaga*.	1) Consumers show positive attitudes, including higher forgiveness, more positive behavioral response, and lower punishment intention, towards a virtual influencer than human influencer. 2) The main effects are consistent in different service failure scenarios including product failure, recommendation failure, and value-based failure.
Pilot-study 2 in Web Appendix	Examining the main effects (H1) again the and testing their generalizability across different influencers' gender.	Pictures of real persons as human influencers and their plotted fictitious images as virtual influencers.	1) The main effects were verified again. 2) The main effects are consistent across different influencers' gender.
Study 1	1). Testing the mediating role of mind perception with its two subdimensions (i.e., agency and experience).2). Excluding alternative explanations of perceived cuteness and perceived novelty.3). Testing the serial mediation model with mind perception and responsibility attribution	The female images as Pilot-study 2.	1) In the two subdimensions of mind perception, perceived agency is a better explanation mechanism than perceived experience. 2) The alternative explanations of perceived cuteness and perceived novelty were excluded. 3) The indirect serial mediation effects of "perceived agency → responsibility attribution" were verified.
Study 2	Testing the moderating role of familiarity.	The real virtual and human influencers on Instagram as experimental materials and their real followers as participants.	1) The familiarity can bring positive effect on consumers' attitudes towards service failures of human influencers but not virtual ones. 2) It is the specific familiarity to certain influencers rather than the general familiarity to the phenomenon of virtual or human influencers plays the moderating roles.

virtual (or social media) influencer on a scale from 1 (not at all familiar) to 7 (very familiar); (2) Their level of fandom towards virtual (or social media) influencers, ranging from 1 (not a fan at all) to 7 (die-hard fan).

Next, the influencer type is manipulated by asking participants in the virtual (or human) influencer condition to view five virtual (or human) influencers' materials. Participants in the familiar (or unfamiliar) condition are then instructed to select the influencer with whom they are most (or least) familiar and respond to subsequent questions. A manipulation check is conducted by asking participants to answer six questions, which include assessments of their familiarity, duration of following, interaction frequency, liking frequency, sharing extent, and commenting frequency with the chosen influencer.

Next, participants are presented with a scenario depicting a service failure by the selected influencer: "*The influencer you've selected above made up a sensational news story about herself to gain more attention and increase her follower count. However, her fans later discovered that the story was untrue.*" Participants are then asked to indicate their forgiveness, behavioral responses, and punishment intention (with reliability indices of Cronbach's $\alpha = 0.91$, Cronbach's $\alpha = 0.79$, and Cronbach's $\alpha = 0.66$, respectively) based on this scenario. The session concludes with participants reporting demographic information and completing attention check items.

5.4. Results and discussions

Manipulation Check. A series of two-way 2 (influencer type) × 2 (familiarity) ANOVAs on the six manipulation check items shows that neither the main effects of influencer type nor the interactions were statistically significant ($Fs \leq 1.15$, $ps \geq .285$), while the main effects of familiarity were all significant ($Fs \geq 7.50$, $ps \geq .007$), suggesting that the manipulation of familiarity is successful.

Main analyses. The two-way 2 (influencer type) × 2 (familiarity) ANOVA on forgiveness propensity reveals a significant interaction effect ($F(1, 287) = 11.63$, $p = .001$, $\eta 2\ p = .04$, see Fig. 4). Simple effects analysis between virtual and human influencer conditions further indicates that the results of Study 1 are replicated in the unfamiliar condition, suggesting that participants show a higher forgiveness propensity towards unfamiliar virtual influencers ($M = 3.29$, $SD = 1.53$, $n = 77$) than unfamiliar human influencers ($M = 2.80$, $SD = 1.27$, $n = 71$, $F(1, 287) = 4.68$, $p = 0.031$, $\eta 2\ p = 0.02$). However, this effect is reversed in the familiar condition, indicating that participants have a higher forgiveness propensity towards familiar human influencers ($M = 3.93$, $SD = 1.00$, $n = 66$) than familiar virtual influencers ($M = 3.23$, $SD = 1.54$, $n = 77$, $F(1, 287) = 9.54$, $p = 0.002$, $\eta 2\ p = 0.03$), showing that familiarity plays a moderating role between the effect of influencer type on consumers' forgiveness propensity. Further simple effects analysis between familiar and unfamiliar conditions reveals that participants have a higher forgiveness propensity towards familiar human influencers than unfamiliar ones ($F(1, 287) = 9.54$, $p = 0.002$, $\eta 2\ p = 0.03$), while such an effect was not significant in virtual influencer groups ($F(1, 287) = 0.09$, $p = 0.769$). Such results suggest that familiarity can improve consumers' forgiveness propensity towards human influencers but not virtual ones. This might be explained by the fact that, compared to virtual influencers, human influencers have certain particularities that make the familiarity with them produce a positive effect on consumers' forgiveness propensity.

Meanwhile, ANOVA results on behavioral responses and punishment intention reveal a similar pattern as those found in forgiveness propensity. Specifically, the interaction effects on behavioral responses ($F(1, 287) = 12.83$, $p < .001$, $\eta 2\ p = .04$, see Fig. 5) and punishment intention ($F(1, 287) = 6.27$, $p = .013$, $\eta 2\ p = .02$, see Fig. 6) are both statistically significant.

Results from simple effects analyses demonstrate that participants tend to have more negative behavioral responses and a higher punishment intention towards unfamiliar human influencers than unfamiliar virtual ones, replicating the results of Study 1. However, such effects disappear in the familiar condition, suggesting familiarity can decrease participants' negative behavioral responses and punishment intention towards human influencers but not virtual influencers (see Web Appendix for statistical details). In conclusion, familiarity can have a certain positive effect on human influencers but not virtual ones in terms of behavioral responses and punishment intention, which is consistent with the results regarding forgiveness propensity.

The above results reveal the moderating role of consumers' familiarity with the specific virtual or human influencer who causes a service failure. Next, we further test whether consumers' general familiarity with the phenomenon of virtual or human influencers can play the same moderating role, using the PROCESS macro in SPSS. We ran six bootstrap moderation analyses (Model 1) in which the scores of the two questions measuring consumers' general familiarity with the phenomenon of virtual or human influencers (i.e., concept familiarity and fandom) served as moderators, and the forgiveness propensity, behavioral response, and punishment intention as dependent variables. As shown in Table 3, all interaction effects are not statistically significant, suggesting that it is the specific familiarity with certain influencers rather than a general familiarity with the phenomenon of virtual or human influencers, can moderate the effect of influencer type on consumers' attitude towards their service failure.

Discussion. Study 2 reveals the moderating role of familiarity in shaping consumer attitudes toward service failure. Specifically, familiarity seems to attenuate consumers' negative attitudes toward human influencers' service failures, but does not alter their attitudes towards

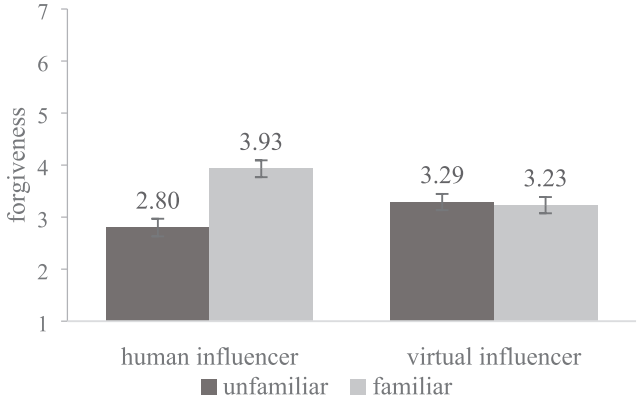


Fig. 4. The interactive effect of influencer type and familiarity on forgiveness propensity.

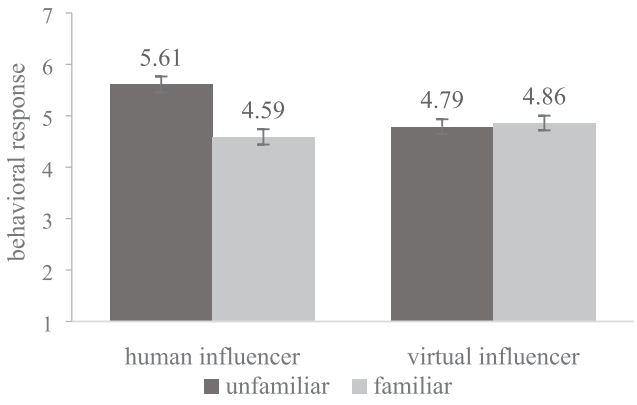


Fig. 5. The interactive effect of influencer type and familiarity on behavioral responses.

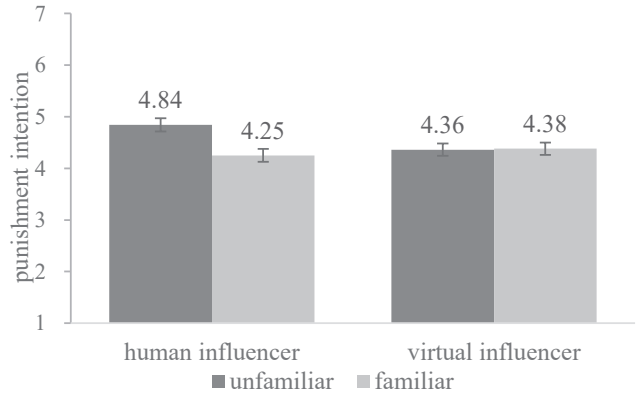


Fig. 6. The interactive effect of influencer type and familiarity on punishment intention.

virtual influencers. This differential effect may be attributed to the unique characteristics possessed by human influencers, facilitating the formation of more robust and personal affection connections with their followers or fans (Bilro and Loureiro, 2023; Fathy et al., 2022; Kwon and Ha, 2023). Such affectionate bonds may lead followers to demonstrate enhanced forgiveness, more positive behavioral responses, and reduced punishment intentions when confronted with service failures. In contrast, the absence of these real human characteristics in virtual influencers may explain why familiarity with them does not exert a similar positive effect. The findings suggest that specific familiarity with a particular influencer, rather than general familiarity with the concept or phenomenon of virtual or social media influencers, serves as the moderating factor, and lend credence to this interpretation. This underscores the significance of the human-like qualities of influencers in shaping the followers' reactions to service failures, emphasizing the distinct roles of human and virtual influencers in this context.

6. General discussion

This study embarked on an in-depth analysis to understand how different types of influencers, specifically virtual and human, influence consumers' responses to service failures. Through two pilot studies and two experimental studies, we uncovered that consumers are more likely to forgive virtual influencers than human influencers during service failures. The mechanisms underlying this effect are found to be rooted in consumers' mind perception and responsibility attribution. The virtual nature of the influencers leads to a lower mind perception, which, in turn, reduces consumers' responsibility attribution and alleviates the negative impact of service failure.

Extending the existing research literature on virtual influencers, the present research reveals the moderating role of familiarity with the influencer that shapes consumer responses. The findings highlight that familiarity can enhance consumers' forgiveness propensity towards human influencers but not virtual ones. This modulation by familiarity adds a new layer of complexity to our understanding of how influencer types interact with consumer psychology.

6.1. Theoretical contributions

Firstly, this study contributes significantly to the evolving field of influencer marketing by articulating a nuanced understanding of the distinctions between virtual and human influencers, particularly in the context of service failures. While previous studies have predominantly centered on influencers' personal characteristics, such as attractiveness, authenticity and credibility (Bi and Zhang, 2023; Chen et al., 2021; Lee and Eastin, 2021; Shukla and Dubey, 2022; Wang and Hu, 2022), this research delves into the psychological mechanisms underlying consumer reactions. By focusing on forgiveness propensity, behavioral responses, and punishment intention, this study reveals the deeper interplay between influencers and consumers, transcending traditional paradigms.

Secondly, the integration of mind perception and responsibility attribution represents a novel theoretical perspective that not only expands but also challenges existing research literature. This study goes beyond merely discerning human from algorithms (Srinivasan & Sarial-Abi, 2021) to explore how mind perception shapes consumers' attribution of responsibility when encountering service failures. The findings shed light on how virtual influencers lead to lower mind perception, which subsequently reduces consumers' responsibility attribution. This intricate relationship advances our understanding of the role of virtual influencers in mitigating the negative impacts of service failure.

Thirdly, the examination of familiarity as a moderating variable provides a significant theoretical contribution. By uncovering how familiarity influences consumers' forgiveness propensity toward human influencers but not toward virtual influencers, this research introduces a new layer of complexity to influencer marketing. Our empirical findings not only deepen the theoretical foundation but also offers practical insights that can guide the crafting of more empathetic and effective communication strategies. The introduction of familiarity as a moderating variable sheds new insights on the boundary conditions of the relationship between influencer type and consumer responses under service failure, providing a complex and multifaceted perspective that has been largely overlooked in existing literature.

Finally, this study transcends the traditional bounds of uncanny valley theory and perceived entertainment (Shin et al., 2019; Gammoh et al., 2018; Dinh & Lee, 2022). By examining consumers' preferences and varied reactions to virtual influencers, the research pioneers a new paradigm in influencer marketing. It not only clarifies the existing theoretical landscape but also carves new paths by exploring previously uncharted territories.

6.2. Managerial implications

Contrary to the intuitive belief that virtual influencers may lack professionalism due to their non-realistic appearance, the findings reveal that virtual influencers can mitigate negative impacts during unexpected service failures. To harness the impact of virtual influencer marketing, digital marketing managers should consider employing unique virtual characteristics for better differentiation and positioning. The deployment of virtual influencers may enhance their effectiveness in reducing negative consumer reactions. This insight urges managers to be cautious about overly humanizing virtual influencers and encourages the exploration of innovative virtual designs.

Firstly, this finding offers a nuanced understanding of how familiarity can enhance consumers' forgiveness propensity towards human influencers but not virtual ones. Managers can strategically leverage this insight to foster more targeted and effective customer engagement strategies, especially in building long-term relationships with consumers (Chen et al., 2022).

Secondly, the insights gleaned from this study also provide valuable guidance for customer-brand relationship repair. By understanding the differential effects of virtual and human influencers and the intricate interplay of mind perception, responsibility attribution, and familiarity, managers can craft more empathetic and responsive strategies. This understanding can aid not only in managing service failures but also in building more resilient and trustworthy customer-brand relationships.

6.3. Limitation and future research

Given the scope of this research focuses on the mechanisms underpinning the relationship between influencer type and consumer response under service failure, there are several limitations warrant further examination. First, the current study has not investigated other

potential boundary conditions that might affect the interpretation of our findings. For example, do the effects observed in our study differ across product categories or consumer segments? Further research should further examine the generalizability of the findings to provide a deeper understanding of which virtual influencers are suitable for different product categories and consumer segments. Second, the stimuli used in the experiments were static, only consisting of images of the influencers and the text description of the service failure scenario. However, in a real-world environment, the interaction between the influencer and the followers is a dynamic and interactive process (Caruelle, 2023; Hamzah and Waqas, 2023). Future research may benefit from enabling real interactions and examining real service failure scenarios in natural or field studies to establish ecological validity. Third, extending from our research, which is mainly based on between-subject experimental designs, future research may also use simulations or other research designs, such as cross-sectional observational studies or time series analyses. Fourth, we recognize the vast potential in further delving into the domain of mind perceptions, especially concerning advertising agents, as this theory is relatively untapped in current literature. As an initial step, researchers might explore how consumers perceive the 'mind' behind newer advertising modalities such as AI-driven personalized advertisements. These cutting-edge advertising methods, driven by advanced technologies, could elicit distinct mind perceptions among consumers. Another intriguing avenue could involve investigating the impact of cultural and demographic differences on mind perceptions, considering the cultural subtleties associated with interpreting intent and emotions. Moreover, as advancements in AI continue to shape the advertising landscape, it would be compelling to study the potential evolution of mind perceptions in the face of increasingly sophisticated AI-driven advertising strategies. Last but not least, further research may employ other types of virtual influencers, such as cartoon or animated figures, to provide additional insights and enrich knowledge development beyond our current findings.

CRediT authorship contribution statement

Taiyang Zhao: Conceptualization, Methodology, Writing – original draft. **Yaxuan Ran:** Conceptualization, Methodology, Writing – original draft. **Banggang Wu:** Conceptualization, Funding acquisition, Resources. **Valerie Lynette Wang:** Formal analysis, Writing – original draft. **Liying Zhou:** Conceptualization, Funding acquisition, Supervision, Writing – original draft, Writing – review & editing. **Cheng Lu Wang:** Writing – review & editing, Supervision, Project administration, Resources.

Data availability

Data will be made available on request.

Acknowledgments

Taiyang Zhao, Yaxuan Ran and Banggang Wu share the first authorship of this paper. Liying Zhou and Cheng Lu Wang share the corresponding authorship of this paper. Please send correspondence to Liying Zhou at zhouly@gufe.edu.cn, or Cheng Lu Wang at cwang@newhaven.edu.

Taiyang Zhao acknowledges the support by the Humanities and Social Sciences Fund Project of Chinese Ministry of Education (Grant 23YJC630250), the National Natural Science Foundation of China (Grant 71902069, 72202084, 72372053, 72272060), and the Fundamental Research Funds for the Central Universities (Grant 2022CXTD16).

Liying Zhou acknowledges the support by the National Natural Science Foundation of China (Grant 72362006), and the Guizhou Provincial Department of Education's Higher Education Scientific Research Project ([2022]176).

Banggang Wu acknowledges the support by the National Natural Science Foundation of China (Grant 71902148, Grant 72372109) and Sichuan University [2023CX33]..

Yaxuan Ran acknowledges the support by the National Natural Science Foundation of China (Grant 72272152).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbusres.2024.114657.

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